

HW 8

To be completed by Wednesday, April 22

1. In this problem you will estimate a logit model using multiple approaches: MLE, NLLS, and MoM. You will examine the determinants of attending a 4-yr college relative to a 2-yr college among post-secondary attendees (this is a repeat of problem 1 from HW2). Use “dataHW8 Problem1.mat”. The data contains an outcome variable Y which is an indicator for attending a 4-yr college. X_i are student observable characteristics including a constant (these coefficients must vary across choices). The second column of X_i is an indicator for whether either parent graduated from a 4-yr college, and the third column of X_i is the student’s high school GPA. Z contains choice specific attributes (the coefficients are common across choices). The first column of Z captures the distance to the closest 2-year college in miles. The second column of Z captures the distance to the closest 4-year college.
 - (a) Estimate a logit model of the 4-yr college choice using **MLE** and X_i and Z as regressors. Report coefficient estimates and SEs.
 - (b) Estimate a logit model of the 4-yr college choice using **NLLS** and X_i and Z as regressors. Report coefficient estimates and bootstrapped SEs.
 - (c) Estimate a logit model of the 4-yr college choice using **MoM** and X_i and Z as regressors. Assume both X_i and Z are exogenous and act as their own instruments. Report coefficient estimates and bootstrapped SEs. Here I recommend using a gradient-free minimizer *fminsearch*, but you can play around with the *fminunc* version.
2. Use “dataHW8 Problem2.mat” to complete this exercise. In this problem you will work with a simple two-period labor supply model with savings. Individuals are endowed with savings in period 1 according to a uniform distribution between $[1.5, 3.5]$. In the first period, individuals will jointly choose whether to work and how much to consume/save. Working yields wages that take on one of five values, $[0.5; 0.75; 1.0; 1.5; 2]$. Note if the observed wage is zero it means that the individual chose not to work. Individuals observe their wage draw prior to making their labor supply choice and you only observe wages for those who choose to work. The true distribution of initial wage draws is determined according to $[\delta_1; \delta_2; \delta_3; \delta_4; 1 - \sum_j \delta_j]$. Wages in the second period are unknown in period one, but transition according to the following:

$$\begin{bmatrix} \rho_1 + \rho_2 + \rho_3 & \rho_2 & \rho_3 & 0 & 0 \\ \rho_2 + \rho_3 & \rho_1 & \rho_2 & \rho_3 & 0 \\ \rho_3 & \rho_2 & \rho_1 & \rho_2 & \rho_3 \\ 0 & \rho_3 & \rho_2 & \rho_1 & \rho_2 + \rho_3 \\ 0 & 0 & \rho_3 & \rho_2 & \rho_1 + \rho_2 + \rho_3 \end{bmatrix}$$

where $\rho_1 + 2 * \rho_2 + 2 * \rho_3 = 1$. In the second (and final) period, individuals again choose whether to work and consume any income and savings.

The flow utility associated with choices c_t, l_t for $t = 1, 2$ is given by

$$u(c_t, l_t) = \beta \ln c_t - \gamma(l_t = 1) + \epsilon_{l_t}$$

where ϵ_{l_t} is an iid Type I EV shock. Note that there are two shocks each period, one associated with working and one associated with not working. Individuals discount the future at a rate equal to 0.9.

The true parameter values are: $\delta_j = 0.2$ for all j , $\rho_1 = 0.4$, $\rho_2 = 0.2$, $\rho_3 = 0.1$, $\beta = 2$, and $\gamma = 0.5$.

- (a) Use dataHW8 Problem2.mat to create a series of moments that you would like to target in estimation. This could include things like the asset distribution in the 2nd period, savings rates, labor force participation rates, average observed wages, initial observed wage distribution, observed wage transitions, etc.

- (b) Simulate data ($N=20,000$) using the actual parameter values and see if your simulated moments are close to the sample moments.
- Create initial conditions drawing wealth and wages in the first period. Construct wage transitions and generate second period wages.
 - Create a grid of savings values and calculate expected utility in period two for every realized wage and savings combination (two state variables).
 - Find optimal consumption in the first period when working and not working. You can do this at their exact wealth level, or you can do this at a set of wealth grid points and interpolate. Note that finding optimal consumption in period 1 requires accounting for future expected utility. Because savings are continuous you will need to interpolate future expected utility between realized wealth grid points in period 2.
 - Using observed utility, calculate the probability of working in the first period and simulate the choice using this probability. Note the form of the probability is driven by the Type I EV error.
 - The realized first period choice determines savings in period 2.
 - Calculate the probability of working in the second period. Note that no consumption decision needs to be determined since individuals consume any income and savings.
 - Generate the same moments you did for the sample.
- (c) Now try and estimate the model parameters by embedding the simulation inside a method of moments style estimator. Be sure to restrict the wage distribution and transition probabilities such that they sum to one. You can do this with a multinomial style transformation inside the estimating routine.